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Women's education and fertility in select countries of Africa and Asia: Moderation by quality of education

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Despite a large literature on education and fertility, most studies are narrowly focused on the level of schooling completed. Does the schooling–fertility relationship depend on the quality of education? Is more schooling still associated with lower fertility even when the quality of education is poor? Drawing on Demographic and Health Survey data from 42 countries in Africa and Asia, this study uses cross-national analysis to examine whether the quality of education moderates the schooling–fertility relationship. Findings show that the strength of the schooling–fertility relationship depends significantly on a country's quality of education: the higher the quality of education, the stronger the negative relationship. But even when educational quality is low, a negative relationship between level of schooling and fertility still exists. By conceptually and empirically distinguishing the level of schooling from the quality of education, this study contributes to a more nuanced understanding of women's education and fertility.

Keywords: fertility; education; learning; cross-national analysis; Demographic and Health Surveys

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Introduction

Demographers have long studied the relationship between women's education and fertility (Cochrane 1979; Cleland and Rodriguez 1988; Jejeebhoy 1995; Martin 1995; Ainsworth et al. 1996; National Research Council 1998; Cleland 2002; Bongaarts 2003; Bongaarts 2010; Shapiro 2013; Lutz and Skirbekk 2014; Glymour and Manly 2018; Hamad et al. 2018; Psaki et al. 2019; Liu and Raftery 2020; Bongaarts and Hodgson 2022; Schoumaker and Sánchez-Páez 2024), but much of the literature is narrowly focused on the duration of schooling or level completed. Spending more years in school does not always lead to more learning. Across low- and middle-income countries, rapid increases in school access and enrolment have not been met with equally rapid improvements in learning outcomes (Pritchett 2013; UNESCO UIS 2017; World Bank 2018; Angrist et al. 2021). It has been estimated that about 617 million children and adolescents worldwide in 2017 did not reach the minimum proficiency in reading and mathematics (UNESCO UIS 2017). A more recent report has indicated that this so-called 'global learning crisis' has worsened, estimating that 70 per cent of 10-year-olds worldwide were unable to understand a simple written text (World Bank et al. 2022).

The acute gap between schooling and learning has been increasingly well documented (Pritchett 2013; UNESCO UIS 2017; World Bank 2018; Angrist et al. 2021). What remains unclear is whether 'schooling without learning' has any consequences for demographic outcomes such as fertility. Some demographers have speculated that low-quality education, as indicated by poor learning outcomes, may have compromised the transformative effect of education on fertility transition (Ainsworth et al. 1996; Grant 2015; Bongaarts et al. 2017; Esteve and Florez-Paredes 2018; Liu and Raftery 2020), while others have argued that poor-quality schooling can still change fertility behaviours and outcomes, even without imparting any relevant knowledge or skills

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(Bledsoe et al. 1998; Basu 2002; Cleland 2002). Despite the existence of a large body of literature on education and fertility, very few studies have distinguished the level of schooling from the quality of education (Glewwe 1998; Psaki et al. 2019), and there is as yet no study directly examining whether the quality of education moderates the relationship between the level of schooling and fertility.

Does the relationship between level of schooling and fertility depend on the quality of education? Is the level of schooling still associated with fertility when the quality of education is poor? In this study, I draw on cross-sectional data from Demographic and Health Surveys (DHS) in 42 countries in Africa and Asia to investigate whether the quality of education moderates the relationship between women's level of schooling and their fertility. I construct a direct country-level measure of educational quality based on the literacy skills of women of childbearing age. Using this measure, I show that the relationship between schooling and fertility is stronger in countries with higher-quality education. However, even when educational quality is low, more schooling is still associated with lower fertility. To explain these findings, I further investigate how the quality of education moderates the relationship between schooling and two proximate determinants of fertility (Bongaarts 1978, 2015): namely, age at first marriage/union and contraceptive use.

By conceptually and empirically distinguishing the level of schooling from the quality of education, this study extends the existing literature on education and fertility. Understanding the moderating role of educational quality helps to explain the varying strengths and natures of the schooling–fertility relationship uncovered by existing studies (Martin 1995; Ainsworth et al. 1996; National Research Council 1998; Bongaarts 2010). It also provides a more nuanced understanding of why and how education affects fertility (National Research Council 1998; Basu 2002; Cleland 2002; Bongaarts 2010; Glymour and Manly 2018; Hamad et al. 2018; Psaki et al. 2019). Moreover, by shedding light on the unintended demographic consequences, findings from this study can contribute to current policy debates on schooling and learning in low- and middle-income countries.

Background: Does the schooling–fertility relationship depend on the quality of education?

The varying relationships between schooling and fertility

Extensive cross-national studies have shown that the relationship between women's schooling and fertility varies across low- and middle-income countries and regions. Cross-sectional analyses using the DHS have demonstrated a pervasive negative association between women's level of schooling and their fertility in recent decades, but large cross-national variation exists in the strength of this association (Martin 1995; Ainsworth et al. 1996; Bongaarts 2003, 2010; Shapiro 2013). Analyses of change over time have found increased schooling to be associated with declines in fertility, although there is much variation across countries in terms of how responsive the fertility trend is to changes in women's educational attainment (Martin 1995; Murtin 2013; Shapiro 2013; Liu and Raftery 2020; Bongaarts and Hodgson 2022; Schoumaker and Sánchez-Páez 2024). Notably, Liu and Raftery (2020) showed that the accelerating effect of increases in women's education on fertility decline was much weaker in sub-Saharan Africa compared with other high-fertility regions. Within sub-Saharan Africa, the rate of fertility decline in some countries is more susceptible to progress in women's education than in others (Schoumaker and Sánchez-Páez 2024).

Similarly, cross-national variation exists in the relationship between women's schooling and some of the proximate determinants of fertility. Although a positive educational gradient in contraceptive use has been found in every DHS country analysed, this educational gradient is much steeper in some countries and regions than in others (Martin 1995; Ainsworth et al. 1996; Bongaarts 2010 Emina et al. 2014; Kantorová and Bongaarts 2024). Substantial cross-national variation also exists in the relationships between women's education and the timing of first birth, first sexual intercourse, and first marriage (Grant 2015; Bongaarts et al. 2017; Esteve and Florez-Paredes 2018). Specifically, evidence from Malawi (Grant 2015) and a number of Latin American countries (Bongaarts et al. 2017; Esteve and Florez-Paredes 2018) suggests that the expansion of school access over time has not led to delays in union formation or childbearing.

This variation in the relationship between education and fertility has led some to speculate that the effect of women's schooling may have been reduced by the low quality of education in certain countries and regions (Ainsworth et al. 1996; Grant 2015; Bongaarts et al. 2017; Esteve and Florez-Paredes 2018; Liu and Raftery 2020). Growing evidence has shown that the rapid increase in school access in many low- and middle-income countries has not always translated into meaningful learning in classrooms (Pritchett 2013; UNESCO UIS 2017; World Bank 2018; Angrist et al. 2021). For

example, in 2017, the UNESCO Institute for Statistics (UIS) estimated that more than 387 million children of primary school age were unable to read with minimum proficiency, and two-thirds of these children were attending school (UNESCO UIS 2017). According to the Programme for International Student Assessment for Development (PISA-D) piloted in seven low- and middle-income countries, even among 15-year-old students who had attained Grade 7 or above, only 23 per cent achieved the minimum level of proficiency in reading and 12 per cent in mathematics (compared with the OECD averages of 80 and 77 per cent, respectively) (Ward 2018). These poor learning outcomes raise concerns about whether the poor quality of education received has prevented individuals from reaping the full benefits of education (World Bank 2018), including the potentially transformative power of education to empower women and improve their lives (Grant 2015).

However, in studies of education and fertility to date, the quality of education, as distinct from the level of schooling, has rarely been considered (Glewwe 1998; Psaki et al. 2019). The narrow focus on level of schooling completed and/or duration of schooling in the existing literature has limited our understanding of the varying strengths and natures of the education–fertility relationship and the likely future course of fertility in low- and middle-income countries as access to education continues to expand without necessarily improving in quality. This study extends the literature by presenting the first moderation analysis of whether and how the schooling–fertility relationship varies by quality of education. Specifically, I ask: is the relationship between level of schooling and women's fertility stronger in countries with higher-quality education? When the quality of education is low, do fertility differentials by level of schooling still exist?

Theoretical perspectives

The potential moderating role of educational quality can be understood through multiple theoretical perspectives on why and how women's schooling affects fertility. According to human capital theory, schooling affects women's fertility by imparting knowledge and skills (Cochrane 1979; Schultz 1988; Bledsoe et al. 1998; Cleland 2002). Empirical research set in a variety of low- and middle-income countries has pointed to literacy as a key skill that explains the relationship between women's schooling and fertility (Oliver 1999; Thomas 1999; Levine et al. 2012). Levine et al. (2012) showed that *academic* literacy (literacy skills taught in school, e.g. reading comprehension) enabled mothers to understand health messages in the print and broadcast media and produce an organized narrative of her own or her child's illness in clinics (Levine et al. 2012). Apart from affecting women's health behaviours, the skills and knowledge acquired with more schooling also increase women's productivity in the labour market (Schultz 1988). Higher wages, and therefore a higher value of women's time, may consequently reduce women's demand for children (Schultz 1997). Whether human capital directly affects women's health behaviours or indirectly affects their demand for children, if the effect of schooling on fertility depends on the knowledge and skills learned in school, then the quality of education is expected to moderate the relationship. Specifically, moderation occurs when the amount of schooling interacts with the quality of an education system to determine the levels of knowledge and skills acquired, which in turn affect fertility. In countries with higher-quality education, the same amount of schooling will lead to higher levels of knowledge and skills, resulting in stronger effects on fertility. Conversely, in countries with low-quality education, more schooling may not translate into any knowledge or skills, and as a result, spending more years in school may not affect women's fertility at all.

Knowledge and skills are neither the only pathway nor a necessary pathway linking schooling to fertility. A set of theoretical propositions in demography, commonly referred to as ideational theory, emphasizes the role of social interaction and the diffusion of new ideas in the fertility transition (Coale and Watkins 1986; Cleland and Wilson 1987; Bongaarts and Watkins 1996; National Research Council 2001). According to this theory, schools play an important role in fertility decline by spreading values and ideas (Cleland and Wilson 1987; Bongaarts and Watkins 1996). Distinct from human capital theory, ideational theory implies that schooling can affect fertility independently of knowledge and skills. In Malawi, Behrman (2015) found that increased schooling led to a decline in women's ideal number of children *without* increasing their engagement with mass media or their knowledge about family planning. Cleland (2002, p. 193) argued that the crucially important link between schooling and fertility is a social-psychological one: schooling lowers fertility by conferring 'a closer identification with modern institutions … and greater confidence in interacting with these institutions'. In both examples, going to school, even without acquiring any knowledge or skills, can reduce fertility. Thus, the ideational

theory of fertility transition implies that the schooling–fertility relationship may not be moderated by the quality of education and that even when education is of poor quality, more schooling may still lead to lower fertility.

Similarly, if more schooling reduces fertility mainly by delaying family formation and childbearing and thus reducing the years of exposure to the risk of pregnancy, the effect may not depend on the quality of education. In most sub-Saharan African countries, going to school is considered incompatible with marriage and childbearing (Lloyd and Mensch 2008). In such contexts, women are expected to have less time or opportunity to have a child while in school, via what Black et al. (2008) called an 'incarceration effect', and those continuing schooling to the next level will have to postpone family formation. The mechanical postponement due to being in school implies no moderation by the quality of education.

Finally, differences in fertility by level of schooling may simply reflect differences in fertility by socio-economic status (Skirbekk 2008; Dribe et al. 2014). If schools mainly reproduce existing social inequality, as posited by a class of sociological theories (Bourdieu 1973; Bowles and Gintis 1976, 2002; Shavit and Blossfeld 1993; Downey and Condron 2016; Jencks 2016), the negative association between level of schooling and fertility will hold regardless of educational quality. In low- and middle-income countries, women's family background may jointly determine their educational attainment and reproductive outcomes (Cochrane 1979, p. 150; Bledsoe et al. 1998, p. 11). Basu (1999, 2002) further demonstrated that in South Asia, men who marry educated women vs uneducated women come from distinct backgrounds, and therefore women's education is also a proxy for husband's background. Empirically, studies on the causal effect of education set in several low- and middle-income countries have revealed that after accounting for socio-economic background using experimental or quasi-experimental designs, the negative effect of education on fertility is greatly reduced in size and, in many cases, no longer statistically significant (Psaki et al. 2019). This suggests that fertility differentials by schooling level may be due largely to differences in socio-economic status rather than the content or quality of what is learned in school.

Taken together, the literature suggests that it is theoretically ambiguous whether the relationship between schooling and fertility is moderated by educational quality. On one hand, if schools impart knowledge and skills crucial for reducing fertility, then when the quality of an education system is poor, the schooling–fertility relationship may be weakened because being in school longer does not result in more knowledge or skills (Ainsworth et al. 1996; Grant 2015; Bongaarts et al. 2017; Esteve and Florez-Paredes 2018; Liu and Raftery 2020). On the other hand, because schooling can lower fertility through multiple pathways that do not depend on what is learned in school (Cochrane 1979, p. 150; Bledsoe et al. 1998; Basu 2002; Cleland 2002), the schooling–fertility relationship may not depend on educational quality at all. The existence of moderation by quality of education thus remains an open question.

Measuring the quality of education

No research to date has empirically tested the moderating role of educational quality in the schooling–fertility relationship. Demographers who study education have long recognized the importance of distinguishing educational attainment from educational quality (National Research Council 1998; Lutz et al. 2008; Grant 2015; Goujon et al. 2016), but a key challenge is the lack of direct, reliable measures of the quality of education that are comparable across countries. Traditionally, cross-national research and policy monitoring have measured the quality of education by 'inputs' into the education system (Tawil et al. 2012), such as the pupil/teacher ratio, availability of textbooks, physical infrastructure, and facilities. However, such inputs often have weak or no effects on student learning and achievement (Hanushek 1995; Glewwe et al. 2011). Recently, there has been an increasing shift to measuring the 'outcomes' of the education system directly by assessing student learning and skills (Tawil et al. 2012; Global Education Monitoring Report Team 2014; World Bank 2018; Angrist et al. 2021; Lutz et al. 2021; World Bank et al. 2022). This has been enabled by the growing number of large-scale international and regional learning assessment programmes conducted in low- and middle-income countries (Ward 2018; Angrist et al. 2021) and by cross-national household surveys that incorporate learning assessments and skills measures (Pierre et al. 2014; UNICEF 2017). Several harmonized data sets have recently been developed to measure and compare learning outcomes and learning-adjusted educational attainment across countries (Tawil et al. 2012; Global Education Monitoring Report Team 2014; World Bank 2018; Angrist et al. 2021; Lutz et al. 2021; World Bank et al. 2022).

Using these latest measures based on learning and skills, several cross-national studies have conducted moderation analyses of educational quality. Pritchett (2024) investigated whether the quality of education moderated the relationship between schooling and economic growth. Using World Bank data on harmonized test scores, he found that the relationship between schooling and economic growth was stronger in countries with higher-quality education: in the Dominican Republic where levels of learning were low, an additional year of schooling added almost nothing to gross domestic product (GDP) per capita, whereas in Singapore where levels of learning were very high, an additional year of schooling added almost $10,000 to GDP per capita (Pritchett 2024). Similarly, Kaffenberger et al. (2018) used moderation analysis to show that the relationship between women's schooling and child survival was stronger in countries with higher-quality education, as measured by the average literacy level of women with five years of schooling. Kaffenberger and Pritchett (2021) argued that equating education with the level of schooling underestimates the real impact of women's education on their life outcomes. Across more than 50 countries, they showed that while women who had completed six years of schooling had fewer births, reported a higher child survival rate, and rated higher on the empowerment index than those who had never been to school, the advantages were even greater for women who had not only completed six years of schooling but could also read (Kaffenberger and Pritchett 2021).

Despite the increasing availability of data on learning and skills, linking the quality of education to demographic outcomes such as fertility poses additional challenges. Cross-national household surveys, such as the Multiple Indicator Cluster Survey (MICS), administer the learning assessment only to children aged below 15, who have yet to reach reproductive age (UNICEF 2017). Household surveys of adult skills (Pierre et al. 2014; OECD 2024) do not collect information on demographic behaviours and outcomes. Large-scale international and regional learning assessments, such as PISA-D (Ward 2018), are limited to testing students of a specific age or grade in school (Angrist et al. 2021), many of whom have yet to reach childbearing age. As a result, even though comparable direct measures of educational quality are increasingly available (Angrist et al. 2021; Lutz et al. 2021; World Bank et al. 2022), data that allow us to link the quality of education received to later reproductive behaviours and fertility outcomes are still scarce.

In this study, I conduct a moderation analysis similar to those of Pritchett (2024) and Kaffenberger et al. (2018), to investigate whether the within-country relationship between schooling and women's fertility varies across countries by the quality of education. To construct a country-level measure of educational quality, I use data from the DHS, which assesses respondents' literacy by administering a simple, standardized reading test during the interview. The literacy skills assessed by the DHS provide a measure of learning that is comparable across countries. The DHS is the only existing data source that allows me to measure the quality of education received by women of reproductive age across countries and to link this directly to their level of schooling and fertility outcomes.

Data and methods

Data

I downloaded data from IPUMS-DHS (Boyle et al. 2024) and pooled the most recent surveys from 42 low- and middle-income countries: 37 in Africa and five in Asia. IPUMS-DHS is an online data dissemination tool that harmonizes data, variables, and codes across DHS surveys over time and across countries, making the use of DHS data more efficient and simpler for cross-national analysis (Boyle et al. 2024). Currently, it covers public data for all DHS countries in Africa and nine countries in Asia (Boyle et al. 2024). From all the available IPUMS-DHS samples, I selected those containing full birth histories for women aged 15–49, women's schooling information, and literacy information measured by asking respondents to read aloud a sentence on a card. This approach of collecting literacy information was used by the DHS in Phase 4 (1997–2003) and later. I excluded earlier DHS surveys because respondents were asked to self-report whether they could read, which is a more biased measure of literacy. If a country had multiple DHS surveys in Phase 4 or later, I used the most recent survey. This gave a pooled sample of 42 DHS surveys representative of 42 countries. Table 1 lists the country, year, and region of the 42 samples that were pooled for analysis. They contain a total of 1,306,625 women aged 15–49 surveyed between 2003 and 2021.

Standardized literacy rate

To measure the quality of education, I constructed a standardized literacy rate for each country. As mentioned earlier, from Phase 4 onwards the DHS asked

Table 1 Description of the 42 DHS surveys pooled for analysis

			Highest level of schooling attended (percentage)				
Region	*Country*	*Year*	*No*	*Primary*	*Secondary or above*	*Literacy rate (percentage)*	*Standardized literacy rate (percentage)*
Western Africa	Benin	2017	55	20	25	30	56
	Burkina Faso	2010	74	14	12	17	58
	Cote d'Ivoire	2011	53	25	21	34	62
	Gambia	2013	46	14	40	42	53
	Ghana	2014	19	18	63	65	53
	Guinea	2018	69	12	20	21	52
	Liberia	2013	33	31	36	40	53
	Mali	2018	66	13	21	24	54
	Niger	2012	80	11	8	11	54
	Nigeria	2018	35	14	51	52	52
	Senegal	2017	46	23	31	40	60
	Sierra Leone	2013	56	14	30	32	52
	Togo	2013	32	33	35	41	54
Southern Africa	Eswatini	2006	8	33	59	86	73
	Lesotho	2014	1	39	60	91	72
	Namibia	2013	5	20	76	90	70
	South Africa	2016	2	9	89	94	68
Northern Africa	Egypt	2014	24	10	66	68	56
	Morocco	2003	50	20	30	46	68
Middle Africa	Angola	2015	22	35	43	51	56
	Cameroon	2018	22	28	51	63	61
	Chad	2014	62	23	15	17	52
	Congo Brazzaville	2011	6	24	70	75	54
	Congo Democratic Republic	2013	15	37	48	55	55
	Gabon	2012	4	21	74	85	62
	Sao Tome and Principe	2008	6	58	36	68	64
Eastern Africa	Burundi	2016	36	39	25	64	75
	Comoros	2012	31	20	49	54	56
	Kenya	2014	7	50	43	80	69
	Madagascar	2021	16	43	42	65	64
	Malawi	2016	12	62	26	64	66
	Mozambique	2011	31	50	19	33	57
	Rwanda	2014	12	64	23	73	70
	Tanzania	2015	15	62	23	73	71
	Uganda	2016	10	57	33	58	62
	Zambia	2018	8	44	48	60	57
	Zimbabwe	2015	1	26	73	91	69
Asia	Bangladesh	2014	25	29	46	54	57
	India	2019	22	12	66	69	55
	Jordan	2017	2	7	91	94	64
	Myanmar	2015	12	41	46	78	71
	Pakistan	2017	49	16	34	46	66

Note: See Standardized literacy rate subsection for details of how the standardized literacy rate is calculated.
Source: Author's analysis of DHS data from 42 countries.

respondents to read aloud a sentence on a card to demonstrate their literacy skill. A respondent was considered literate if she could read a whole sentence or illiterate if she could read only part of the sentence or cannot read at all. Table 1 lists the literacy rate (i.e. proportion of women who can read a whole sentence) in each country; this ranges from 11 per cent in Niger to 94 per cent in South Africa. While the literacy rate can be used to measure a country's educational quality directly, one issue is that it is highly correlated with level of schooling. As illustrated in Figure 1, there is a strong positive

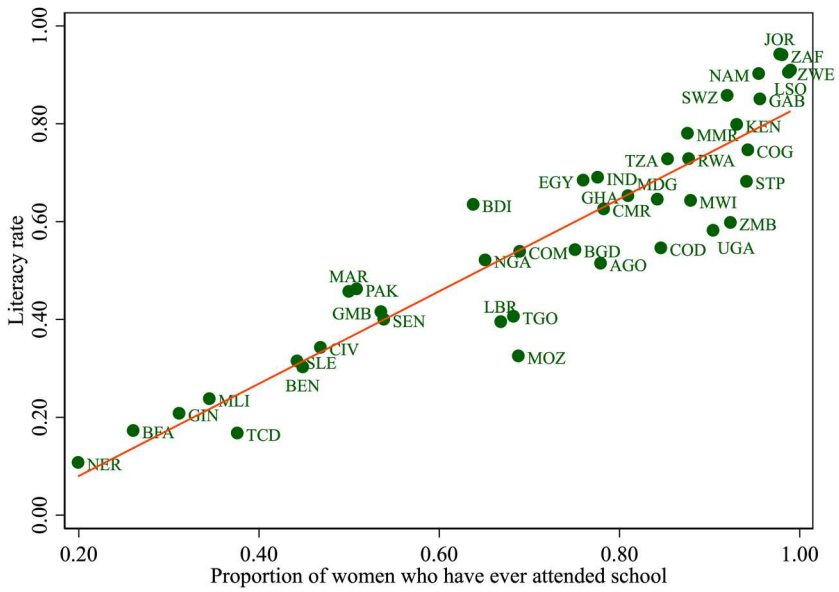

Figure 1 Scatter plot of the level of schooling and unadjusted literacy rate in 42 African and Asian countries
Source: Author's analysis of DHS data from 42 countries.

correlation between the proportion of women who have ever attended school and the proportion who are literate. The correlation coefficient is 0.92, indicating that the proportion ever attending school and literacy rates are measuring the same construct. In other words, the literacy rate is a poor measure of a country's quality of education because it is indistinguishable from the country's level of schooling.

Instead, this paper uses the *standardized* literacy rate as a measure of educational quality that adjusts for differences in the level of schooling. For each country, I first calculated a literacy rate specific to each level of schooling. I then calculated an average literacy rate assuming a uniform distribution of highest schooling level attended for all countries (i.e. 25 per cent no schooling, 25 per cent primary, 25 per cent secondary, and 25 per cent tertiary). For instance, without standardization, Zimbabwe's literacy rate was 91 per cent in 2015, much higher than in Pakistan, where the literacy rate was 46 per cent in 2017 (Table 1). However, a closer look at the schooling composition of the two countries reveals that Zimbabwe's level of schooling is much higher than Pakistan's: 1 per cent of women in Zimbabwe have never attended school vs 49 per cent in Pakistan (Table 1). This means that the disparity in literacy rates is likely due to the disparity in schooling access between the two countries rather than their disparate quality of education per se. Indeed, once a uniform distribution of schooling levels is applied, the standardized literacy rates for the two countries are much more similar (69 per cent in Zimbabwe vs 66 per cent in Pakistan). Table 1 lists the standardized literacy rate for each country. Across the 42 countries sampled, the standardized literacy rate ranges from 52 to 75 per cent, with a mean of 61 per cent and a standard deviation of 0.07.

A limitation of the DHS literacy measure is that in earlier phases women who had attended secondary or higher education were assumed to be literate without being asked to read the passage. This means that the quality of education is likely to have been overestimated (Sandefur 2017). It was not until Phase 7 (2013–18) that the DHS started measuring the literacy skills of those who had attended secondary education (The DHS Program 2017). In my pooled sample, 18 surveys measured the literacy skills of those who had attended secondary education or higher, and 24 surveys assumed that all women with secondary or higher education were literate. To keep the standardized literacy rate measure comparable across countries, in the main analysis presented here I follow the earlier DHS approach, assuming all women with secondary or higher education are literate. In the sensitivity analysis presented later in the paper, I address this limitation in more detail and show that my results are robust to whether secondary-educated women are assumed to be literate.

Model specifications

The main analysis investigates whether the within-country relationship between the level of schooling and the total fertility rate (TFR) varies across

countries by quality of education. To model the relationship between schooling and the TFR, I first transformed the full birth histories collected by the DHS and computed, for each woman, her number of births and exposure to risk of giving birth within the three years preceding the survey date (Moultrie 2013; Croft et al. 2023). Following the latest guide to DHS statistics (Croft et al. 2023) and *Tools for Demographic Estimation* (Moultrie 2013), data for the three years preceding the survey date were used to calculate period fertility rates because rates for a single calendar year derived from survey data may not be reliable. I then grouped the person-age data by one-year age group (from 15 to 49), level of schooling (no schooling, primary schooling, or secondary schooling or higher), place of residence (rural or urban), and country. For each group, I aggregated the number of births and person-years of exposure (Moultrie 2013; Croft et al. 2023). The grouped data set contains the number of births and exposure for 8,820 groups (defined by 35 one-year age groups, three levels of schooling, two types of residence, and 42 countries), of which 8,578 have non-zero exposure.

The TFR can be estimated by fitting a Poisson model to the grouped data set with exposure as an offset and an indicator variable for age (Schoumaker 2013, p. 1111). (Poisson regression can also be fitted to the person-age data set without grouping, which is equivalent to fitting the Poisson regression to the grouped data [Schoumaker 2013, p. 1099]. However, using grouped data greatly reduces data storage needs and computation time compared with using person-age data.) To model the relationship between level of schooling and fertility, I added a variable indicating level of schooling to the Poisson model, and the coefficients tell us the schooling differences in TFR expressed as rate ratios (Schoumaker 2013, p. 1113). I also controlled for rural/urban residence because it might confound the relationship between schooling and fertility. Most importantly, because my analytic sample is pooled from DHS surveys conducted in different countries, I fitted a Poisson model with country-level fixed effects (Allison 2009; Cameron and Trivedi 2013, p. 351). Formally, for each group defined by age i, schooling level j, and place of residence k within country s, let μ_{ijks} denote the aggregated number of births in that group and n_{ijks} the exposure in person-years. The Poisson fixed-effect model can be specified as:

$$\log \mu_{ijks} = \log n_{ijks} + \alpha_i + \beta_j + \gamma_k + \delta_s \qquad (1)$$

As mentioned earlier, the exposure, $\log n_{ijks}$, which enters the model as an offset, and the indicator of one-year age group, α_i, form the basic Poisson model for estimating fertility (Schoumaker 2013, p. 1111). The highest level of schooling attended is indicated by β_j (with the omitted category 'no schooling'). Thus, the exponentiated coefficients, $\exp(\beta_j)$, represent the TFR ratios relative to no schooling (Schoumaker 2013, p. 1113). The model also controls for a dummy variable of rural/urban residence, γ_k (with the omitted category 'urban'). The country fixed effects, δ_s, effectively control for any country-level heterogeneity including but not limited to socio-economic development, phase of fertility transition, and access to family planning.

In equation (1), the schooling–fertility relationship, represented by the exponentiated coefficients, $\exp(\beta_j)$, is assumed to be the same in all countries. However, the goal of this paper is to test whether the schooling–fertility relationship varies across countries by quality of education, so I extended the model defined in equation (1) by letting the schooling–fertility relationship depend on a country's standardized literacy rate. Let x_s denote the standardized literacy rate of country s centred around the mean. The extended model can be specified as:

$$\log \mu_{ijks} = \log n_{ijks} + \alpha_i + \beta_j + \beta_j x_s + \gamma_k + \delta_s \qquad (2)$$

Compared with equation (1), equation (2) adds an interaction term between level of schooling and the standardized literacy rate: $\beta_j x_s$. Because the country fixed effects, δ_s, already control for any country-level variation, it is not necessary or possible to include the main effect of the standardized literacy rate, x_s, which varies across countries, so only the interaction term between the level of schooling (which varies within countries) and standardized literacy rate needs to be specified (Allison 2009). With x_s centred around the mean, the exponentiated coefficients for the level of schooling, $\exp(\beta_j)$, now represent the fertility rate ratios relative to no schooling *when the quality of education is average*. The interaction term, $\beta_j x_s$, which is what we are most interested in, tells us whether and how the quality of education moderates the association between the level of schooling and fertility.

Proximate determinants of fertility

To understand the moderation of the schooling–fertility relationship further, in this paper I also examine whether the quality of education moderates the relationship between schooling and two proximate determinants of fertility: age at first marriage/

union and contraceptive use. The proximate determinants of fertility are a set of biological and behavioural factors that directly influence fertility (Bongaarts 1978). If any background variable (e.g. education) affects fertility, it necessarily affects one or more of the proximate determinants of fertility (Bongaarts 1978). Therefore, if the quality of education moderates the relationship between schooling and fertility, it should also moderate the relationship between schooling and one or more of the proximate determinants. In this paper, I focus on marriage/union exposure and contraception (Bongaarts 1978, 2015) because previous studies have shown that these two proximate determinants are the key explanations for fertility differentials by educational level, especially in sub-Saharan African countries (Bongaarts et al. 1984; Singh et al. 1985; Jolly and Gribble 1993; Bongaarts 2010).

The sample for analysing the proximate determinants of fertility contains women aged 15–49. Table 2 summarizes the variables used to measure marriage/union exposure and contraception. At the time of survey, 74 per cent of all women in the analytic sample have ever entered a marriage/union, and 70 per cent have ever had a birth. To study the prevalence of contraception, we can examine the proportion of women who are currently using any contraceptive method. In the analytic sample, among women who are currently married or in a union, 14 per cent are currently using any contraceptive method. To understand contraceptive use further, I measured whether the woman had a need for family planning and whether that need was satisfied, using standard indicators defined by the DHS. These DHS indicators are derived and constructed from women's responses to questions about their fecundity, fertility preferences, and use of contraceptives. The DHS measure considers a woman to have a need for family planning if she is currently using any contraceptive method or if she is assumed to have an unmet need for family planning (Croft et al. 2023, p. 7.42). Among women with a need for family planning, the second DHS measure considers the need satisfied if the woman is currently using any contraceptive method (Croft et al. 2023, p. 7.42). Further details on the indicator definitions and variable constructions are available in Bradley et al. (2012) and the guide to DHS statistics (Croft et al. 2023, p. 7.42). Although the DHS measures of need for family planning, especially the assignment of unmet need, have been subject to criticism (Bradley and Casterline 2014; Senderowicz et al. 2023), they are widely used in academic studies and policy monitoring. On applying these definitions to the analytic sample, I find that 59 per cent of women who are currently married or in a union have a need for family planning, of which 85 per cent have their need satisfied.

To model ages at first marriage/union and first birth, I fitted parametric survival models where the survival time is assumed to follow a log-normal distribution. To model contraceptive use, need for

Table 2 Measures of the proximate determinants of fertility used in the analysis

	Mean/ proportion	*Number of women*
Marriage/union exposure[1]		
Ever entered a marriage/union	0.74 (0.44)	1,234,606
Age at first marriage/union (among those who have ever entered a marriage/union)	18.80 (4.25)	912,428
Ever had a birth	0.70 (0.46)	1,222,408
Age at first birth (among those who ever had a birth)	20.19 (3.96)	854,665
Contraception[2]		
Currently using any contraception	0.14 (0.35)	902,821
Having a need for family planning	0.59 (0.49)	902,821
Need for family planning satisfied (among those with a need)	0.85 (0.36)	533,760

[1]Excluding four DHS surveys that sampled only ever-married women (Bangladesh 2014, Egypt 2014, Jordan 2017, and Pakistan 2017).
[2]Among women who are currently married or in a union.
Note: Standard deviations in parentheses.
Source: As for Table 1.

family planning, and whether the need was satisfied, I used linear probability models. Unlike for the TFR, for the proximate determinants I fitted models to pooled individual-level data for women aged 15–49. In each model, I used interaction terms between levels of schooling and the standardized literacy rate to test moderation by quality of education, and I controlled for age at time of survey, rural/urban residence, and country fixed effects.

Results

Schooling and fertility: Moderation by quality of education

Model 1 in Table 3 presents the exponentiated coefficients from fitting the Poisson model specified in equation (2). As explained in the Data and methods section, the exponentiated coefficients for levels of schooling should be interpreted as fertility rate ratios relative to having no schooling. Thus, according to Model 1, when the quality of education is average and controlling for rural/urban residence, the TFR for women who have attended primary school is 8 per cent lower than for those with no schooling, and for women who have attended secondary school or higher, the TFR is 29 per cent lower than for those with no schooling. Our key interest is moderation by quality of education, which is indicated by the exponentiated coefficients for the two interaction terms between the level of schooling and quality of education (as measured by the standardized literacy rate). Both are less than 1.0, indicating that the negative association between levels of schooling and the TFR is stronger when the quality of education is higher. The Chi-square statistic (Table 3) from a joint significant test indicates that the relationship between women's level of schooling and the TFR varies significantly by the quality of education.

Therefore, Model 1 (Table 3) confirms that the quality of education, as measured by the standardized literacy rate, does indeed moderate the relationship between level of schooling and fertility and that the relationship is stronger in countries with higher-quality education. What remains unclear is whether a negative association between schooling and fertility still exists when educational quality is low. In other words, is 'schooling without learning' still associated with lower fertility? To further probe the moderation effect (Hayes 2017), Figure 2 uses the estimated coefficients in Model 1 (Table 3) to visualize the fertility differentials by level of

Table 3 Moderation of schooling–fertility relationship by quality of education: Exponentiated coefficients estimated from the Poisson model specified in equation (2)

	All surveys *(1)*	*Surveys sampling all women* *(2)*	*Asia* *(3)*	*Africa* *(4)*
Level of schooling (omitted category: No schooling)				
Primary	0.918**	0.903***	0.804***	0.914***
	(0.024)	(0.022)	(0.004)	(0.018)
Secondary or higher	0.711***	0.680***	0.622***	0.645***
	(0.022)	(0.009)	(0.008)	(0.013)
Interactions (omitted category: No schooling × Quality)				
Primary × Quality	0.544*	0.449**	0.321***	0.457*
	(0.159)	(0.129)	(0.010)	(0.149)
Secondary or higher × Quality	0.152**	0.111***	0.052***	0.328***
	(0.096)	(0.063)	(0.002)	(0.094)
Residence (omitted category: Urban)				
Rural	1.310***	1.320***	1.258***	1.296***
	(0.019)	(0.019)	(0.005)	(0.027)
Chi-square statistic	9.84**	20.90***	1283.13***	16.85***
Number of groups	8,578	7,740	420	7,320

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

Notes: Standard errors (in parentheses) are clustered at the country level. All models control for one-year age groups and country fixed effects and apply DHS individual sampling weights for women. Model 1 covers all 42 countries, whereas Models 2–4 exclude DHS surveys that sampled only ever-married women (Bangladesh 2014, Egypt 2014, Jordan 2017, and Pakistan 2017). The Chi-square statistic tests the hypothesis that the coefficients for the interaction terms are jointly zero.

Source: As for Table 1.

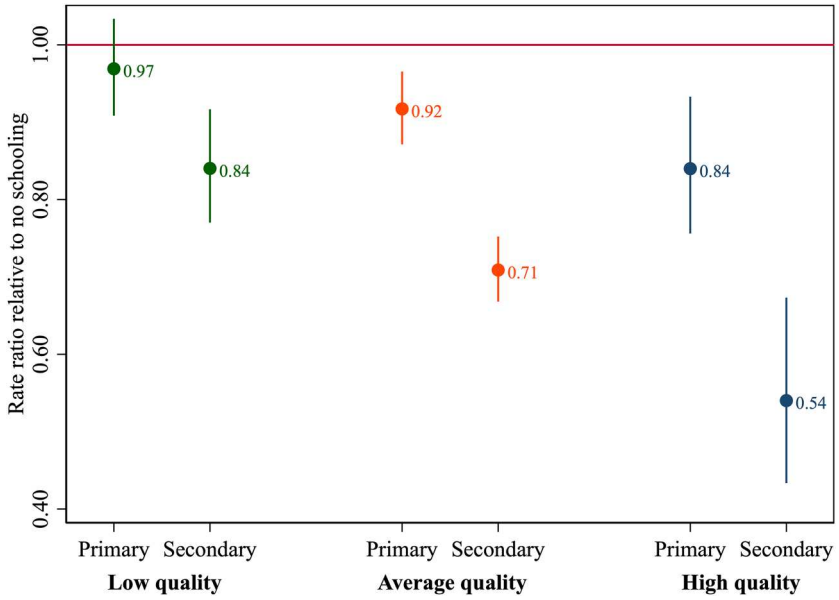

Figure 2 Relationship between level of schooling and the TFR when the quality of education is low, average, and high: Rate ratios relative to women with no schooling

Notes: Rate ratios are calculated using estimates reported in Model 1, Table 3. Vertical lines show 95 per cent confidence intervals. Low-, average-, and high-quality education are defined as standardized literacy rates at the sample minimum (52 per cent), mean (61 per cent), and maximum (75 per cent), respectively.

Source: As for Figure 1.

schooling when the quality of education is low, average, and high, as indicated by the sample minimum (52 per cent), mean (61 per cent), and maximum (75 per cent) of the standardized literacy rate, respectively. Two things stand out from Figure 2. First, the higher the quality of education, the steeper the negative schooling gradient in TFR. Second, even when the quality of education is low, a negative and significant schooling gradient in TFR still exists.

Four of the countries pooled in my analysis sampled only ever-married women in their DHS, which could have biased my results. For Model 2 (Table 3) I removed these four countries from the sample and repeated the same analysis. The results are very similar. In Models 3 and 4, I conducted the same analysis separately for Asian and African countries, and the same conclusion holds within each region: moderation by quality of education is statistically significant, and the negative association between schooling and fertility is stronger in countries with higher-quality education.

Schooling and proximate determinants of fertility: Moderation by quality of education

If quality of education moderates the relationship between schooling and fertility, it should also moderate the relationship between schooling and at least one of the proximate determinants of fertility. In Model 1, Table 4, I present the estimated coefficients from a parametric survival model assuming that age at first marriage/union follows a log-normal distribution. The positive and significant coefficients on schooling levels indicate that when holding rural/urban residence constant, women with a higher level of schooling enter first marriage/union later. However, the interaction terms between levels of schooling and quality of education are not statistically significant, meaning that the relationships between higher levels of schooling and later first marriage/union do not depend on the quality of education. In other words, variation in educational quality across countries does not moderate the relationship between schooling and age at first marriage/union. The results are very similar when examining age at first birth (Model 2, Table 4).

Next, from Model 3 (Table 4), we can examine the prevalence of contraception among women who are currently married or in a union. The coefficients were estimated from a linear probability model of whether a woman is currently using any contraceptive method. When quality of education is average and controlling for rural/urban residence, women who have attended primary school or secondary school are about seven and nine percentage points,

Table 4 Moderation of the relationship between schooling and the proximate determinants of fertility by quality of education: Estimated coefficients for levels of schooling and interactions between levels of schooling and quality of education (measured by the standardized literacy rate)

	Exposure to marriage/union		*Contraception*		
	Age at first marriage/ union (1)	*Age at first birth (2)*	*Use (3)*	*Have a need (4)*	*Need satisfied (5)*
Level of schooling (omitted category: No schooling)					
Primary	0.048***	0.020***	0.067**	0.057***	0.061***
	(0.008)	(0.005)	(0.022)	(0.010)	(0.015)
Secondary or higher	0.197***	0.143***	0.090***	0.041**	0.089***
	(0.008)	(0.004)	(0.024)	(0.012)	(0.019)
Interactions (omitted category: No schooling × Quality)					
Primary × Quality	0.137	0.088	0.788*	0.175	0.689*
	(0.114)	(0.084)	(0.380)	(0.219)	(0.254)
Secondary or higher × Quality	0.135	0.152	1.068*	0.447	1.019**
	(0.159)	(0.114)	(0.446)	(0.349)	(0.358)
Residence (omitted category: Urban)					
Rural	−0.045***	−0.035***	−0.026	−0.031*	−0.031
	(0.004)	(0.003)	(0.014)	(0.012)	(0.019)
Chi-square statistic	1.55	1.79	3.25*	1.69	4.05*
Number of women	1,222,408	1,222,408	902,821	902,821	533,760

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.
Notes: Columns (1) and (2) report coefficients from survival models assuming survival time follows a log-normal distribution. Columns (3)–(5) report coefficients from linear probability models. Standard errors (in parentheses) are clustered at country level. All models control for country fixed effects. Models 3–5 control for age at time of survey. Models 3–5 control for age at time of survey. The Chi-square statistic tests the hypothesis that the coefficients for the interaction terms are jointly zero.
Source: As for Table 1.

respectively, more likely to use contraception than those with no schooling. Moreover, the interaction terms between schooling and the standardized literacy rate are positive and statistically significant, meaning that the relationship between schooling and contraceptive use is significantly stronger when the quality of education is higher. Using the estimated coefficients, Figure 3 illustrates the differential probabilities of contraceptive use by level of schooling when educational quality is low, average, and high. Notably, when quality of education is low, schooling makes no difference to the prevalence of contraception. It is only when quality of education is at an average or high level that contraceptive use differs significantly between women who have attended primary or secondary school and those with no schooling.

Models 4 and 5 (Table 4) further investigate whether the moderating role of educational quality is driven by the need for family planning or the extent to which that need is satisfied. The estimated coefficients for the interaction terms indicate that the positive association between level of schooling and having a need for family planning is not moderated by the quality of education (Model 4). In contrast, the quality of education significantly moderates the relationship between schooling and whether the need for family planning is satisfied (Model 5).

My earlier analysis revealed that the negative association between schooling level and fertility is stronger when quality of education is higher (see Figure 2). The analysis of the proximate determinants of fertility further reveals that moderation by quality of education works through contraception. Specifically, when the quality of education is high, contraceptive use is more prevalent among women with schooling, whereas when the quality of education is low, schooling does not make a difference to contraceptive prevalence. However, I find that quality of education does not moderate the relationship between schooling and timing of first marriage/union or first birth: more schooling postpones the age at first marriage/union and first birth to the same extent regardless of educational quality.

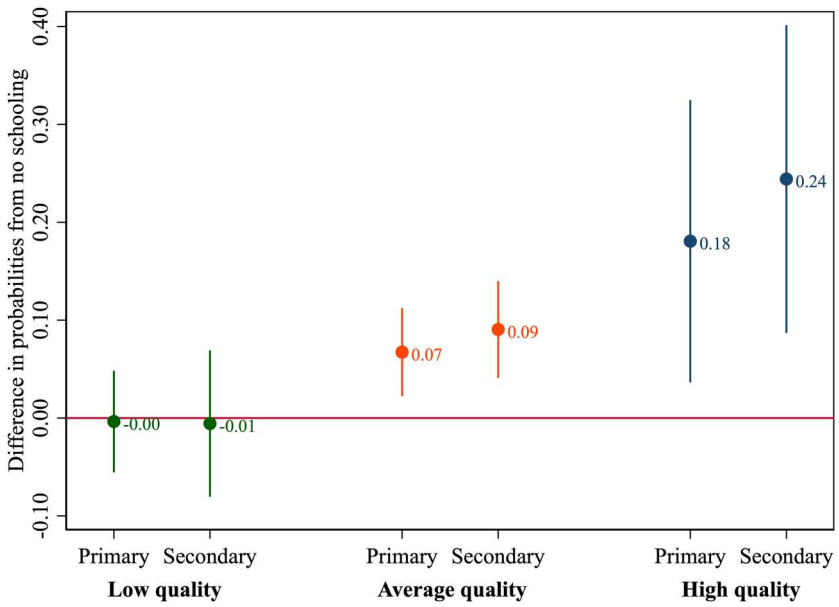

Figure 3 Relationship between level of schooling and the prevalence of contraception when educational quality is low, average, and high: Differences in probabilities of using any contraceptive method compared with women with no schooling

Notes: Differences in probabilities are calculated using estimates reported in Model 3, Table 4. Vertical lines show 95 per cent confidence intervals. Low-, average-, and high-quality education are defined as standardized literacy rates at the sample minimum (52 per cent), mean (61 per cent), and maximum (75 per cent), respectively.

Source: As for Figure 1.

Sensitivity analysis

Alternative model specifications

My main analysis shows that the relationship between the level of schooling and the TFR depends significantly on the quality of education. Other country-level characteristics that are correlated with the quality of education may also modify the schooling–fertility relationship. In Appendix Table A1, I present models with additional controls. One concern is that the schooling gradient in TFR may change as low- and middle-income countries enter the later stages of the fertility transition (Bongaarts 2003). Model 2 thus extends the model specified in equation (2) by letting the coefficients on schooling levels vary by a country's stage of fertility transition (i.e. TFR > 5, TFR = 3–5, and TFR < 3). Even after accounting for the stages of fertility transition, the estimated coefficients for the interaction terms between schooling and standardized literacy rate remain largely unchanged (Model 2, Table A1), and the schooling differences in TFR still vary significantly by quality of education. Another concern is that because my pooled sample contains surveys spanning 2003–21, moderation by quality of education may be confounded by change over time. Model 3 (Table A1) controls for interactions between schooling levels and survey year. There is little change in the estimated coefficients for the interactions between schooling levels and quality of education.

Alternative measures of educational quality

As mentioned previously, one limitation of using the DHS literacy measure is that most of the earlier DHS surveys assumed that all women who had attended secondary school or higher were literate. To keep the quality of education measure comparable across samples, the analysis so far has followed this approach in assuming all women with secondary or higher education to be literate. This means that the quality of education may have been overestimated (Sandefur 2017). Next, I check if my results are biased by this measurement error. First, I used a subsample of 18 DHS surveys from Phase 7 or later, which measured the actual literacy skills of women with secondary school education instead of assuming them to be literate (The DHS Program 2017). For each of these 18 surveys, I calculated two standardized literacy rates, one assuming all women with secondary education to be literate and the other using their actual literacy skills. Appendix Figure A1 presents a scatter plot of the two

standardized literacy rates. The correlation coefficient is 0.99, indicating an almost perfect correlation between the two rates. This suggests that assuming women with secondary education to be literate should not have biased the measurement of country-level quality of education much.

In the main analysis, I standardized the literacy rate by applying a uniform distribution of schooling levels to all countries. To check the robustness of my results, I instead applied the schooling-level composition of Lesotho in 2014 (1 per cent no schooling, 38 per cent primary, 52 per cent secondary, and 9 per cent tertiary) to standardize literacy rates across countries. Model 4 (Table A1) shows that using this alternative standard does not change my results much.

Moreover, my analyses so far have defined and measured quality of education at the country level. Previous research has suggested measuring quality differences within countries (Grant 2015). While the DHS does not contain school-level information, I can define quality of education at subnational level. Specifically, I calculated a standardized literacy rate for each DHS domain (the subnational geographic unit for which representative estimates of indicators can be produced; ICF International 2012). Then I estimated a similar model to equation (2) but with domain-level instead of country-level fixed effects. The domain-level analysis, presented in Model 5 (Table A1), generates a very similar conclusion as the country-level analysis: the negative schooling–fertility relationship is significantly stronger in subregions with higher-quality education.

Finally, I checked if my quality-of-education measure using the DHS literacy question correlates with alternative country-level measures of educational quality. I focused on eight African countries in my sample that participated in both DHS Phase 7 (2013–18) and the Southern and Eastern Africa Consortium for Monitoring Educational Quality (SACMEQ III), a cross-national learning assessment of literacy and numeracy skills among Grade 6 students in 2007. Other countries in my sample either did not have alternative learning assessment data available or had learning assessment data only for younger cohorts of women who had yet to reach reproductive age and thus were not covered by the DHS (World Bank 2022). I calculated the standardized literacy rate among women aged 15–25 in the DHS (who would have been of school age when SACMEQ was administered roughly 10 years earlier) and compared it with the 'learning deprivation gap', a country-level learning outcome measure calculated using the SACMEQ literacy assessment data (World Bank 2022). While I could not directly link participants in the DHS and SACMEQ, the two measures of country-level quality of learning should be highly correlated at least. Appendix Figure A2 confirms that among the eight African countries, the standardized literacy rate based on the DHS (used in my main analysis) is indeed highly correlated with the learning outcome measure based on SACMEQ, with a correlation coefficient of 0.81. This reassures us of the validity of my measure of educational quality using the standardized literacy rate.

Completed fertility

The main analysis focused on the TFR, a hypothetical measure summarizing the age-specific fertility rates of a particular *period*. Here, I take a cohort perspective and measure completed fertility using the number of children ever born to women aged 40–49 (who have reached the end of their reproductive years). If moderation by quality of education were mainly through changing the timing of childbearing rather than the number of births, I would expect the results to be very different when using completed fertility vs the period TFR.

Using a subsample of women aged 40–49, I first recalculated the standardized literacy rate for each country and then fitted a Poisson model with country-level fixed effects to estimate the number of children ever born. Appendix Table A2 presents the exponentiated coefficients from the estimated Poisson model, and Appendix Figure A3 uses these to illustrate the schooling gradient in completed fertility when educational quality is low, average, and high. Overall, the quality of education still significantly moderates the strength of the relationship between schooling and completed fertility: the negative schooling gradient in completed fertility is steeper as quality of education increases. The Chi-square statistic (reported in Table A2) from a joint significant test indicates that the relationship between women's level of schooling and completed fertility varies significantly by the quality of education. Compared with the period TFR, the slight difference when using completed fertility is that the association between primary education and completed fertility is no longer moderated by quality of education. Taken together, my results are robust to using completed fertility and consistent with my analyses of the proximate determinants of fertility: while quality of education changes the age patterns of

fertility, moderation by quality of education cannot be accounted for by the tempo effect alone.

Discussion and conclusions

Across low- and middle-income countries, the rapid expansion in school access has not been met with similar progress in learning (Pritchett 2013; UNESCO UIS 2017; World Bank 2018; Angrist et al. 2021). As concerns rise about the consequences of poor learning outcomes, the policy debate is also starting to shift away from how to increase school access to how to improve learning (Beeharry 2021; Center for Global Development 2021; Evans and Hares 2021; World Bank 2023). Absent from this debate, however, is a clear understanding of the demographic consequences of the acute gap between schooling and learning. Results from this study show that the relationship between level of schooling and women's fertility depends significantly on the quality of education in a country: on average, the higher quality the education, the stronger the negative relationship between level of schooling and fertility. However, even when the quality of education is low, significant fertility differentials by schooling level exist.

Why is the relationship between schooling and fertility stronger in countries with higher-quality education? The study reveals that this is because the differences in contraceptive use by schooling level rely crucially on the quality of education. On average, in countries where educational quality is low, schooling does not make a difference to the prevalence of contraceptive use among women. Only in countries with average- or high-quality education is contraceptive use more prevalent among women who have been to school than those who have not. This finding conforms to previous studies on literacy skills and women's health behaviours (Oliver 1999; Thomas 1999; Levine et al. 2012) and can be interpreted through human capital theory. In countries with low-quality education, more schooling may not translate into the knowledge and skills that enable women to meet their contraceptive needs. This could help to explain why educational quality moderates the strength of the relationship between schooling and fertility. By contrast, the schooling differences in the timing of first marriage/union/birth do not depend on educational quality. Across countries, more schooling is associated with later age at first marriage/union/birth, but the association does not vary significantly by quality of education. This is consistent with what Black et al. (2008) termed the incarceration effect of education, whereby being in school alone reduces women's exposure to the risk of pregnancy. That more schooling delays family formation and childbearing regardless of the quality of education offers one explanation for why a negative schooling gradient in fertility exists even when the education is of low quality.

This study has several limitations. First, my analytic sample does not cover all low- and middle-income countries. It does not cover regions other than Asia and Africa and includes only five countries in Asia. Second, the analysis is focused on the cross-sectional relationship, comparing the fertility of women with different levels of schooling in a given period. Consequently, the findings cannot be extrapolated to relationships between changes in education and fertility decline across cohorts or over time. Finally, due to the data limitations and empirical challenges discussed previously, I rely on the DHS literacy question to measure the quality of education at country level. As more cross-national learning assessment data become available and the current cohorts of students reach reproductive age, future research will be better able to capture other dimensions of learning and more valid measures of educational quality at national and local levels can be devised.

Despite these limitations, this study presents the first evidence that the quality of education moderates the relationship between level of schooling and women's fertility. This moderation effect suggests that the variation in educational quality across low- and middle-income countries provides a plausible explanation for the varying strengths of the schooling–fertility relationships observed in previous studies (Martin 1995; Ainsworth et al. 1996; Bongaarts 2003, 2010; Shapiro 2013). It also supports the hypothesis that poor-quality education may have attenuated the effects of increased schooling on fertility decline in certain countries and regions (Ainsworth et al. 1996; Grant 2015; Bongaarts et al. 2017; Esteve and Florez-Paredes 2018; Liu and Raftery 2020). More specifically, findings from this study reveal that poor-quality education weakens the schooling–fertility relationship through moderating the relationship between schooling and contraception: when educational quality is poor, women with more schooling are no more likely to use contraception or satisfy their need for family planning than those with less schooling.

Moreover, the study sheds light on the demographic consequences of 'schooling without learning' in low- and middle-income countries. Even though

poor-quality education may weaken the effect of women's schooling on fertility, it does not eliminate it. As this study demonstrates, on average, even when the quality of education is low, differences in fertility by schooling level still exist. This suggests that high-quality education is not a necessary condition for more schooling to lead to changes in fertility behaviours and outcomes, echoing the arguments of previous studies (Bledsoe et al. 1998; Basu 2002; Cleland 2002). As Basu (2002, p. 1780) aptly noted, 'there is something about the *fact* of going to school or being schooled, independently of what school itself teaches, that must account for at least a part of the highly changed individual that the schooled girl becomes.' Future research should continue to explore the distinct roles of schooling vs learning and investigate the alternative pathways linking schooling to fertility.

This improved understanding of the moderating role of educational quality has important policy implications. Increases in women's educational attainment have been shown to be a dominant socio-economic driver of fertility transition (Murtin 2013; Liu and Raftery 2020; Bongaarts and Hodgson 2022). Although access to schooling continues to expand across low- and middle-income countries, there is rising concern over the stagnation and even deterioration in educational quality (Pritchett 2013; UNESCO UIS 2017; World Bank 2018; Angrist et al. 2021). Will the continued rise in women's educational attainment lead to further changes in fertility if learning outcomes remain poor? This study suggests that continued expansion of schooling access may still influence women's fertility behaviours and outcomes, but poor-quality education may attenuate the effect. Meanwhile, as the global education community considers shifting the policy priority towards foundational learning (Beeharry 2021; Center for Global Development 2021; Evans and Hares 2021; World Bank 2023), the findings of this study suggest that improving foundational skills, especially literacy, could have the unintended effect of bolstering the benefits of education for women's demographic outcomes.

Notes and acknowledgements

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Appendix

Table A1 Moderation of schooling–fertility relationship by quality of education: Exponentiated coefficients estimated from the Poisson model specified in equation (2) with additional controls and alternative measures of educational quality

	All surveys (1)	*Controlling for Stage of fertility transition (2)*	*Controlling for Survey year (3)*	*Alternative schooling composition (4)*	*Domain-level educational quality (5)*
Level of schooling (omitted category: No schooling)					
Primary	0.918**	0.916**	0.877*	0.916**	0.992
	(0.024)	(0.028)	(0.049)	(0.026)	(0.005)
Secondary or higher	0.711***	0.606***	0.483***	0.707***	0.842***
	(0.022)	(0.032)	(0.074)	(0.024)	(0.004)
Interactions (omitted category: No schooling × Quality)					
Primary × Quality	0.544*	0.507*	0.569	0.616*	0.724***
	(0.159)	(0.145)	(0.169)	(0.135)	(0.044)
Secondary or higher × Quality	0.152**	0.222**	0.234*	0.238**	0.309***
	(0.096)	(0.108)	(0.141)	(0.113)	(0.021)
Residence (omitted category: Urban)					
Rural	1.310***	1.292***	1.302***	1.309***	1.321***
	(0.019)	(0.018)	(0.018)	−(0.018)	−(0.015)
Chi-square statistic	9.84**	10.78**	6.32*	9.93**	365.64***
Number of groups	8,578	8,578	8,578	8,578	342,803

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.
Notes: Standard errors (in parentheses) are clustered at country level in Models 1–4 and at domain level in Model 5. All models control for one-year age groups. Models 1–4 control for country fixed effects. Model 1 replicates Model 1 of Table 3. Model 2 controls for interactions between schooling levels and stages of the fertility transition. Model 3 controls for interactions between schooling levels and survey year (centred around 2003). Model 4 uses an alternative schooling composition to calculate the standardized literacy rate (i.e. 1 per cent no schooling, 38 per cent primary, 52 per cent secondary, and 9 per cent tertiary). Model 5 uses the standardized literacy rate defined at the DHS domain level and controls for domain fixed effects. The Chi-square statistic tests the hypothesis that the coefficients on the interaction terms are jointly zero.
Source: Author's analysis of DHS data from 42 countries.

Table A2 Moderation of the relationship between schooling and completed fertility: Exponentiated coefficients estimated from a Poisson model of number of children ever born to women aged 40–49

	Completed fertility
Level of schooling (omitted category: No schooling)	
Primary	0.877***
	(0.016)
Secondary or higher	0.656***
	(0.011)
Interactions (omitted category: No schooling × Quality)	
Primary × Quality	1.032
	(0.271)
Secondary or higher × Quality	0.500*
	(0.168)
Residence (omitted category: Urban)	
Rural	1.141***
	(0.022)
Chi-square statistic	11.29**
Number of individuals	253,345

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.
Notes: Standard errors (in parentheses) are clustered at the country level. The model controls for country fixed effects, applies DHS individual sampling weights for women, and excludes DHS surveys that sampled only ever-married women (Bangladesh 2014, Egypt 2014, Jordan 2017, and Pakistan 2017). The Chi-square statistic tests the hypothesis that the coefficients on the interaction terms are jointly zero.
Source: Author's analysis of DHS data from 38 countries.

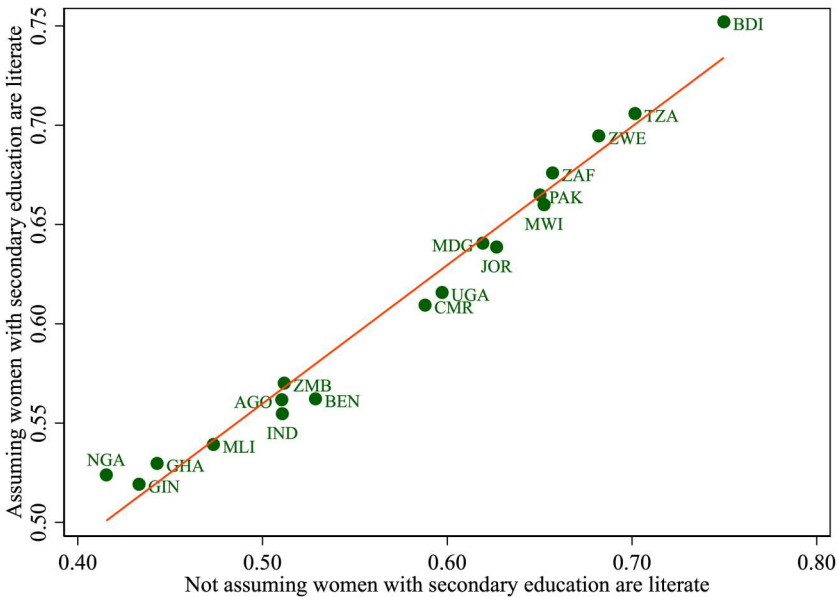

Figure A1 Scatter plot of standardized literacy rate assuming women with secondary education to be literate vs using their actual literacy skills
Source: Author's analysis of DHS data from 18 countries (where actual literacy skills were measured).

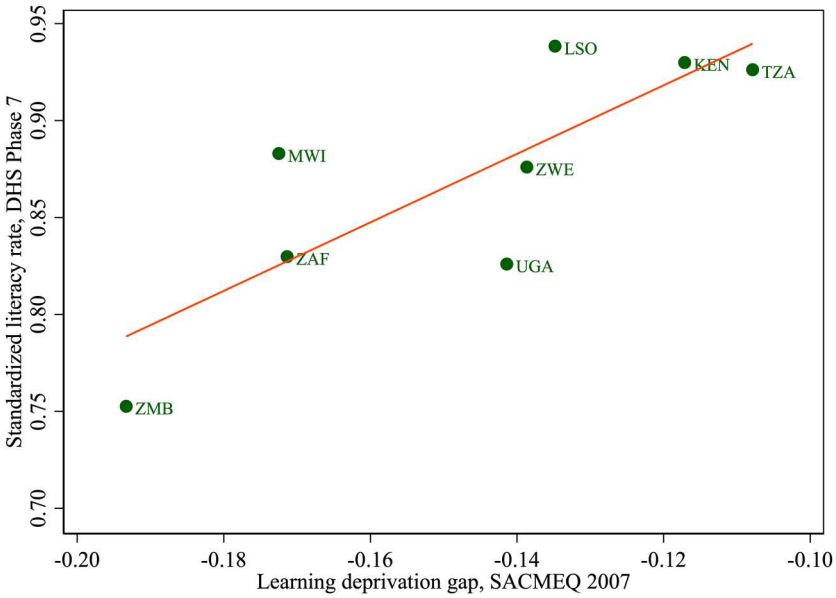

Figure A2 Scatter plot of standardized literacy rate using DHS data vs learning deprivation gap using SACMEQ 2007 data: Eight African countries
Source: Author's analysis of DHS data from eight countries and the Global Dataset of Country Learning Outcome (World Bank 2022).

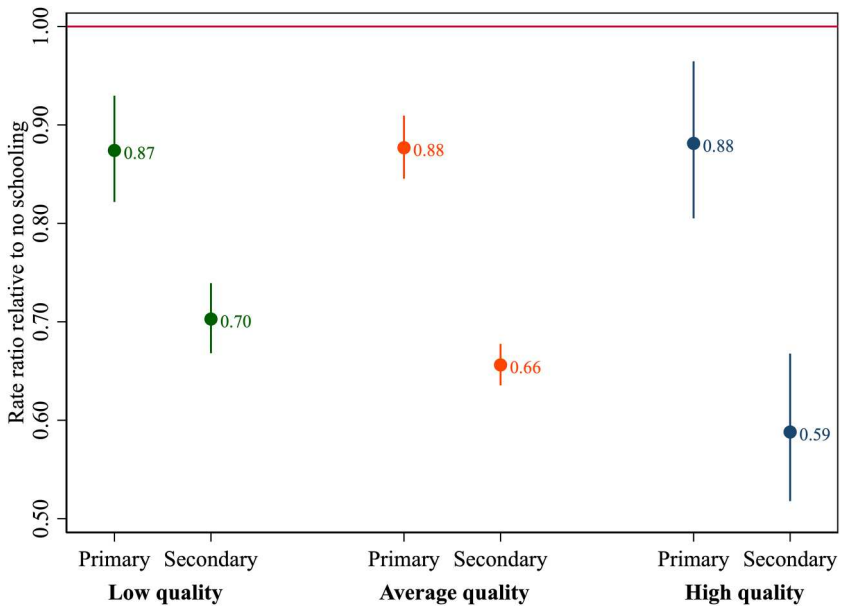

Figure A3 Relationship between level of schooling and completed fertility at low, average, and high quality of education: Rate ratios relative to women with no schooling

Notes: Rate ratios are calculated using estimates reported in Model 1, Table A2. Vertical lines show 95 per cent confidence intervals. Low quality, average quality, and high quality are defined as standardized literacy rates at the sample minimum (51 per cent), mean (61 per cent), and maximum (77 per cent), respectively.

Source: Author's analysis of DHS data from 38 countries.